**KOMALPREET SINGH**
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## SUMMARY

Remote Sensing & Geospatial Data scientist (Ph.D.) who builds machine learning systems on satellite data, from raw sensor files to deployed tools. Built a global cloud detection pipeline across 6 geostationary satellites, generated foundation-model training data at scale (457K patches), and deployed a web-based cholera early-warning system used for 4-week forecasts. Currently developing hourly air-quality prediction models across the US with PyTorch and gradient boosting on Sentinel, MODIS, VIIRS and reanalysis data.

## SKILLS

- **Machine learning & Deep learning:** PyTorch, scikit-learn, XGBoost, LightGBM, TabNet, TabPFN, Temporal Fusion Transformer, CNN segmentation (CSDNet, OmniCloudMask), Bayesian modeling, Optuna, SHAP, OpenCV
- **Remote Sensing & Geospatial:** Google Earth Engine, Microsoft Planetary Computer, Satpy, GDAL, Rasterio, Xarray, NetCDF/GRIB, GeoTIFF/COG, Zarr, QGIS, ArcGIS Pro/ArcPy, geemap, leafmap, IRIS; Sentinel-1/2, MODIS, VIIRS, GOES, Himawari, MSG, MTG, PACE, GPM
- **Atmospheric & Earth system modeling:** WRF-ARW/WPS, ERA5, HRRR, GEOS-CF, MERRA-2, plume and dispersion modeling, atmospheric correction, column density retrieval
- **Infrastructure:** Python, R, MATLAB, Linux/HPC, Docker, AWS CLI, Cloudflare, Git/GitLab, Weights & Biases, Streamlit, Tableau, Power BI

## PROFESSIONAL EXPERIENCE

**Postdoctoral Research Fellow | UTHealth Houston** **Sep 2026 – Present**

- Building geospatial data pipelines using Google Earth Engine and Microsoft Planetary Computer with integrated satellite remote sensing (Sentinel-1/2, MODIS, VIIRS), reanalysis and chemical-transport-model data (ERA5, HRRR, GEOS-CF), emission inventories with physics-based plume modelling, and EPA/TCEQ monitoring data.
- Developing spatiotemporal machine learning models in Python (PyTorch, LightGBM, XGBoost, TabPFN, Temporal Fusion Transformer) to predict hourly air quality across the US. The work involved Optuna hyperparameter tuning, SHAP interpretation, ensembling, and leakage-free spatial, temporal and random cross-validation.

**Remote Sensing Scientist | Satlantis US** **May 2025 – June 2026**

**Global Cloud Detection**

- Built an automated cloud detection system across 6 geostationary satellites (GOES-18, GOES-19, Himawari-8, Himawari-9, MSG-2, MTG-I1) covering the full Earth disk, producing weighted cloud statistics and GeoJSON exports for global weather analysis.
- Fixed an MSG-2 GRIB2 coordinate extraction failure by implementing EUMETSAT's inverse geostationary projection, restoring sub-kilometer geolocation for Indian Ocean monitoring.
- Fine-tuned OmniCloudMask on 3,256 images labeled with the IRIS annotation tool and applied it to 5 TB+ of imagery.

**Foundation-Model Training**

- Generated 457,491 training patches (~2 TB) at L1D and L1G levels for PAN and VNIR bands across multiple sensors.
- Created 15,000 Sentinel-2 L2A RGB-NIR patches for land-use/land-cover training of the Clay 1.5 foundation model.

**Methane plume detection**

- Built a CSDNet cloud-segmentation pre-filter that reduced cloud-induced false positives in methane plume detection and improved processing efficiency by 23%.
- Developed a physics-based feature extraction and OpenCV segmentation workflow to isolate methane plumes from WRF concentration fields, increasing processing speed by 30%.
- Developed and validated two independent methods for quantifying methane point-source emissions against ground-truth measurements.
- Streamlined Git workflows, reducing merge conflicts by 30% for collaborative version control across team of 5 researchers.

**Graduate Research Assistant | University of Florida** **Jan 2022 - Dec 2025**

- Built & deployed CholDSS, a Streamlit web application for satellite-based cholera risk detection, cutting a 6-hour manual workflow to under 3 minutes, to predict 4-week advance cholera risk forecasts globally. Cholera Risk Assesment
- Designed a TabNet deep learning model for Vibrio suitability across 350,506 locations from 11 oceanographic variables (accuracy 94%, AUC 0.943).
- Mapped global Vibrio habitat suitability for summer 2025 with XGBoost across 450,786 spatial points (accuracy 84.7%).
- Processed 10 TB+ of MODIS, GPM, MERRA-2, PACE and TRMM data into analysis-ready inputs to identify environmental drivers of Vibrio dynamics.
- Analyzed 30+ coastal water samples in the lab for Vibrio detection and quantification, linking field data to satellite observations.

**NSF Intern | Florida Department of Environmental Protection** **Aug 2023 - Dec 2023**

- Conducted comprehensive oceanographic field collection across 20+ sites, precisely measuring parameters including temperature, salinity, pH, and dissolved oxygen (±0.2°C, ±1%, ±0.2 pH, ±0.1 mg/L DO).
- Optimized in-situ sampling workflows, reducing post-collection processing time by 25% and enhancing real-time field capture and validation.
- Implemented precise in-situ measurement techniques using YSI ProDSS, achieving a 98% completeness rate and enhancing the integrity and reliability of information for downstream environmental analysis.

**Graduate Research Scholar | Punjab Agricultural University** **Aug 2019 - Sep 2021**

- Developed a Python (Tkinter) spatial decision support system for irrigation management, giving water volume (cusecs) and pump capacity (HP) recommendations for 530,265 villages of Punjab, India with an accuracy of 84.72%.
- Built a data pipeline combining remotely sensed precipitation, air and land surface temperature, solar radiation and groundwater levels to compute village-scale evapotranspiration, backed by a spatial database of all villages.
- Evaluated satellite-based navigation systems for tractor guidance, improving operational efficiency by 45.2% and reducing overlap and coverage gaps (published 2021).

## EDUCATION

Ph.D. | Environmental Health Engineering | University of Florida | GPA:3.91/4.00 Jan 2022 – Dec 2025
M.S. | Soil and Water Engineering | Punjab Agricultural University | GPA: 7.5/10 Aug 2019 – Sep 2021
B.S. | Agricultural Engineering| Punjab Agricultural University | GPA: 7.5/10 Aug 2015 – Jun 2019

## PUBLICATIONS

**Peer-reviewed journal articles**

- Jamal, Y., Usmani, M., Brumfield, K. D., **Singh, K.**, Huq, A., Nguyen, T. H., Colwell, R., & Jutla, A. (2024). Quantification of climate footprints of Vibrio vulnificus in coastal human communities of the United States Gulf Coast. *GeoHealth*. [https://doi.org/10.1029/2023GH001005]
- Brumfield, K. D., Usmani, M., Santiago, S., **Singh, K.**, Gangwar, M., Hasan, N. A., Netherland, M., Jr., Deliz, K., Angelini, C., Beatty, N. L., Huq, A., Jutla, A. S., & Colwell, R. R. (2023). Genomic diversity of Vibrio spp. and metagenomic analysis of pathogens in Florida Gulf coastal waters following Hurricane Ian. *mBio*. [https://doi.org/10.1128/mbio.01476-23]
- Jutla, A., Usmani, M., Brumfield, K. D., **Singh, K.**, McBean, F., Potter, A., Gutierrez, A., Gama, S., Huq, A., & Colwell, R. R. (2023). Anticipatory decision-making for cholera in Malawi. *mBio*. [https://doi.org/10.1128/mbio.00529-23]
- **Singh, K.**, Prakash, A., Singh, M., Sharda, S., & Gupta, H. (2021). Relative performance of satellite-based navigation systems for improving tractor productivity. *Agricultural Research Journal*. [DOI No. 10.5958/2395-146X.2021.00125.3]

**Under review**

- **Singh, K.**, Usmani, M., Kumar, S., Brumfield, K. D., Colwell, R. R., & Jutla, A. (2026). Remote Sensing to Predict Climate-Sensitive Pathogens: a Review. *GeoHealth*.
- **Singh, K.**, Brumfield, K. D., Colwell, R. R., & Jutla, A. (2026). Predictive Geospatial Intelligence for Cholera: An Operational Decision Support System for Early Warning and Resource Allocation. *International Journal of Health Geographics.*
- **Singh, K.**, Magers, B. M., Brumfield, K. D., Colwell, R. R., & Jutla, A. (2026). Applying Machine Learning Algorithms to Understand the Seasonal Risk of Vibrio spp. in Coastal Waters. *Environmental Health Perspectives.*
- Magers, B. M., Nguyen T., Lipp E., **Singh, K.**, Revere L., Glasser J., Brumfield, K. D., Loboda T., Usmani M., Beatty N., Colwell, R. R., Lakshmi V., & Jutla, A. (2026). Climate, Health, and National Security: A Need for a Centralized Biothreat Intelligence and Early Warning Framework for Environmentally Sensitive Pathogens. *Environmental Health Perspectives.*
- Kamau L., Sese E., **Singh K.**, Kumar S., Jutla A., & Akanda A. (2026). Satellite-Based Geospatial Cholera Early Warning and Smartphone Dissemination in Kenya. *GeoHealth.*